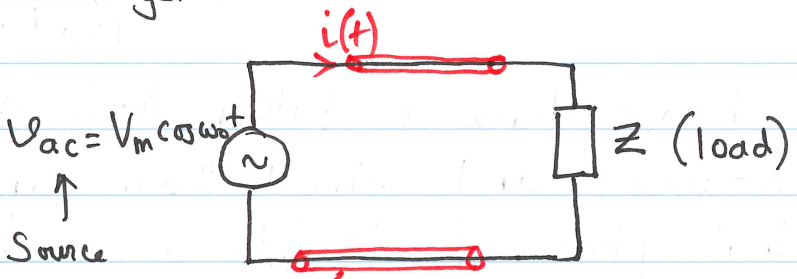
Lecture 5.

Suggested Reading – Section (2.5 – 2.7) Pai.

Last time – * Power factor correction
– importance and how to approach.
* Introduced nomenclatures in three phase power systems.

Today: * Why three phase?
* Generic approach to attack three phase problems?

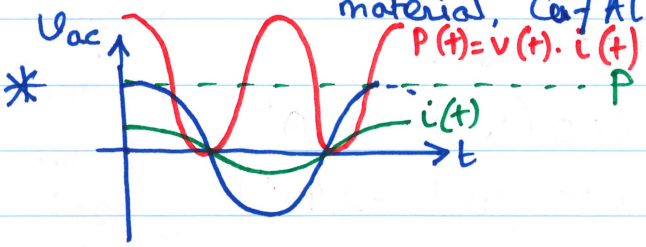
Let us step back and re-look at our single-phase power system:



Whats the objective – transfer power from A to B?

[But also economically]

cost of conducting material, Cu / Al



Three challenges:
1. Power is oscillatory
2. For trasferring 'P' amount of power from A to B, need two cables – Cu, Al
– insulators.

3. Once we convert this p(t) (electrical) to mechanical, oscillatory torque/speed. [For light-is ok].

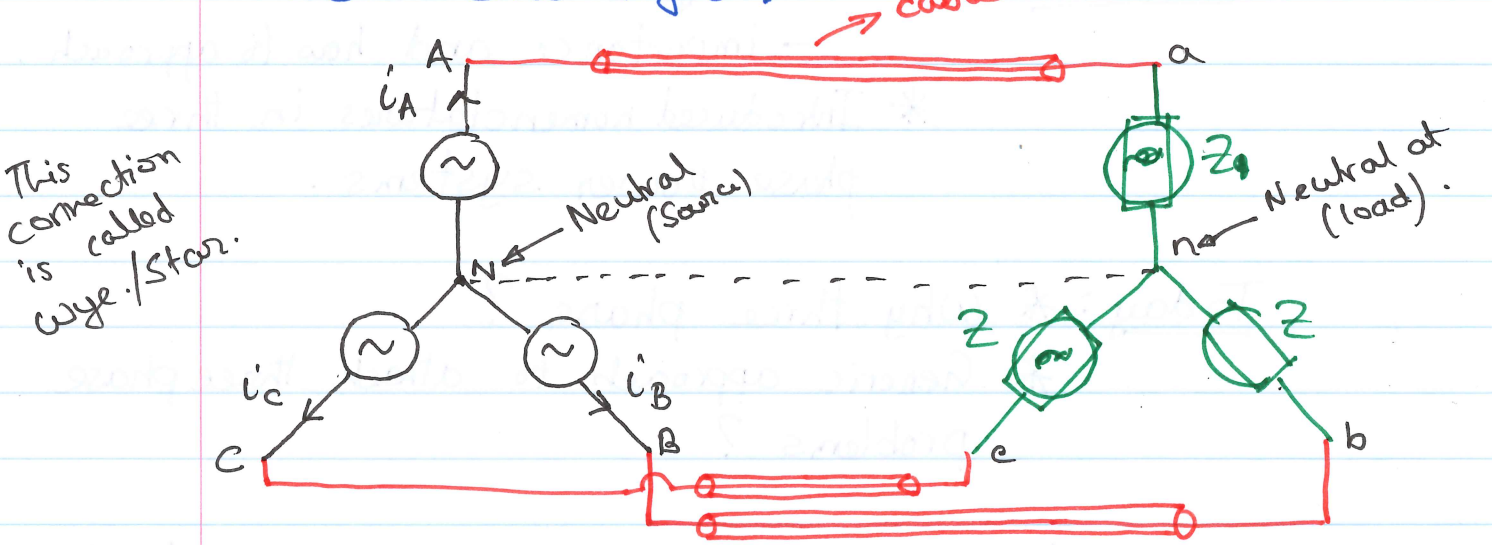
PLEASE remember:

v_{AN}, v_{BN}, v_{CN} = phase voltage at source

v_{AB}, v_{BC}, v_{CA} = line voltage at source

i_A, i_B, i_C = phase / line current.

Q. How three-phase power system solves these challenges?



Three-phase "balanced" source

Three-phase "balanced" load. → all equal impedances.

All we have done is taken three singhe phase sources and wired them up with three single-phase loads (in a special way). However, these single-phase sources are "special" ← "Balanced".

$$v_{AN}(t) = V_m \cos \omega_0 t$$
$$v_{BN}(t) = V_m \cos(\omega_0 t - 120^\circ)$$
$$v_{CN}(t) = V_m \cos(\omega_0 t - 240^\circ)$$

} same magnitude. same frequency phase shifted by 120°.

$\therefore$ $i_A(t) = I_m \cos(\omega_0 t - \phi)$ ← As if a single phase load. with Z impedance.

$$i_B(t) = I_m \cos(\omega_0 t - 120^\circ - \phi)$$
$$i_C(t) = I_m \cos(\omega_0 t - 240^\circ - \phi)$$

** $i_N(t) = i_{AN}(t) + i_{BN}(t) + i_{CN}(t)$

$= 0 \quad \forall t$ · [i.e no need to connect 'n' and 'N']

As the sources are "balanced", driving a "balanced" set of loads, the currents are "balanced" as well.

↓

i.e, same magnitude, same frequency, phase shifted by 120° relative to each other.

Let us compute instantaneous ~~total~~ power.

$$p(t) = v_{AN}(t) \cdot i_A(t) + v_{BN}(t) \cdot i_B(t) + v_{CN}(t) \cdot i_C(t)$$

⋮

lot of trig identies ["Good luck" :-)].

$$= \frac{3}{2} V_m I_m \cos\phi \cdot$$ [independent of time.]

$$= 3 V_{rms} I_{rms} \cos\phi \cdot$$ implying constant power for all time.

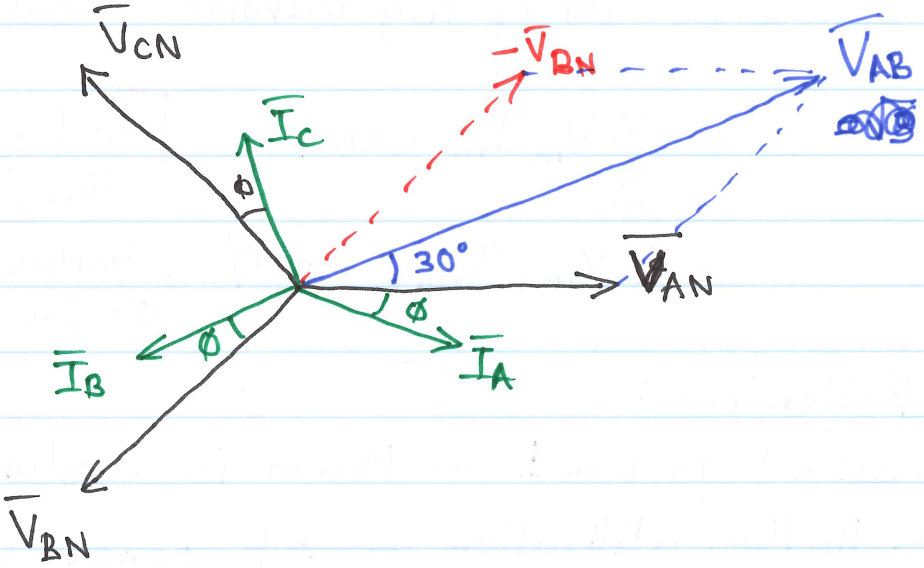
Bottomline:

1. Solved problem 1 — Power is constant over time.
2. Better utilization of infrastructure — 3P power using [3] cables [compared to 1 power wig 2 cables]. ** why
3. If you convert this power ~~to~~ to mechanical, torque/speed can be both constant. "NICE".

CAVEAT: HERE EVERYTHING NICE AND BALANCED, REAL-WORLD - IMPOSSIBLE, measures taken to make as balanced as possible.

* Also, three phase power makes more sense at bulk power transmission level. /distribution level.
* At locality /house, single phase is okay. How do we create balance then, as we convert to single-phase distribution? THINK.

Q. How do we extend our concepts of phasors in three phase



Q. How do we convert phase voltages to line voltages or vice versa?

Line voltage phasor $\leftarrow \bar{V}_{AB} = \bar{V}_{AN} - \bar{V}_{BN}$

$= \sqrt{3}\,|\bar{V}_{AN}| \angle 30° \leftarrow$ leading by 30°

↑ magnitude scaled by $\sqrt{3}$.

∴ Active power per phase

$$= |\bar{V}_{AN}||\bar{I}_A| \cos\phi \cdot$$

Three phase (balanced)

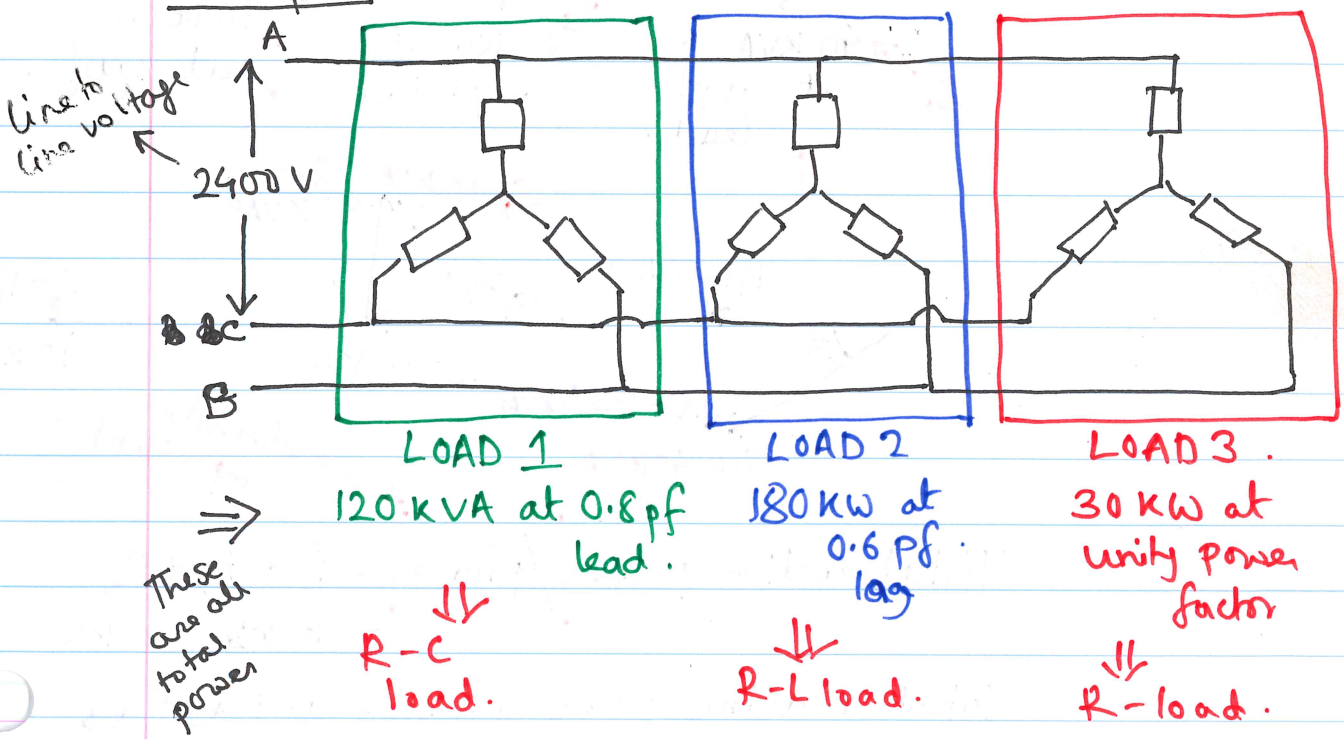
Total active power, $P = 3|\bar{V}_{AN}||\bar{I}_A| \cos\phi$

$$= \sqrt{3}|\bar{V}_{AB}||\bar{I}_A| \cos\phi \text{ W.}$$

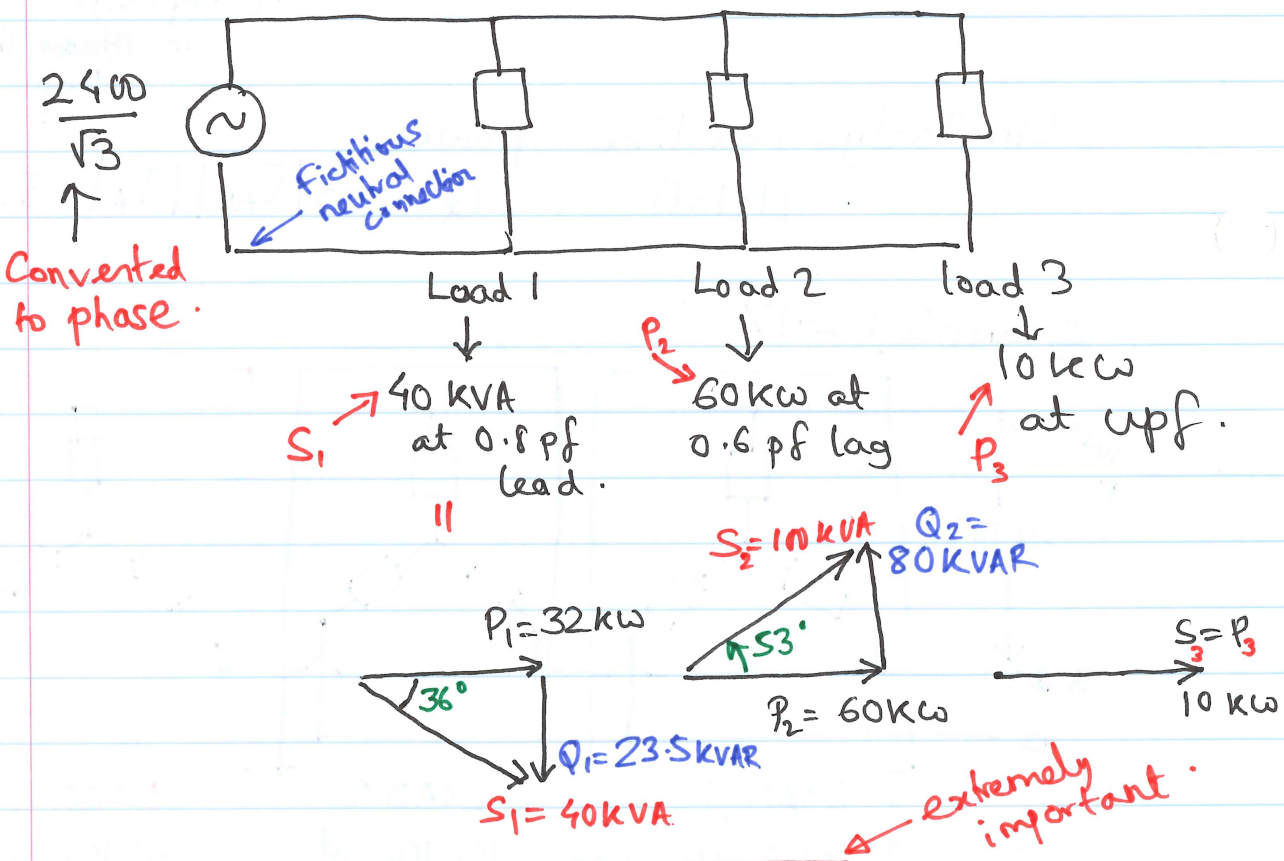
↑ if expressed in ~~phase~~ line voltage.

Similarly, reactive power (total), $Q = 3|\bar{V}_{AN}||\bar{I}_A| \sin\phi \cdot$

Example: 2.17



* You know how to solve single-phase circuits.
* Goal: Reconstruct single phase circuits (valid because source/load all are assumed to be balanced).
* Solve the problem in single phase
* Go back to 3-phase quantities, if required.



extremely important.

Total active power/phase $= P_1 + P_2 + P_3$

$= 102$ kw

Total reactive power/phase $= Q_1 + Q_2 + Q_3$

$= 56.5$ KVAR.

(a) total complex power :

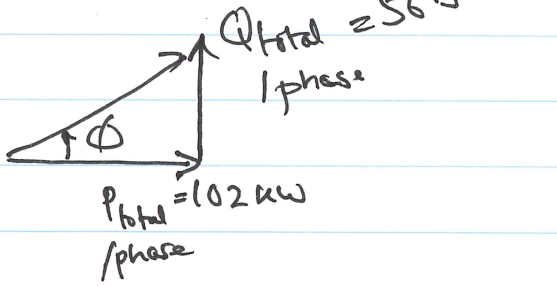
extremely important.

$$\bar{S}_T = 3(P + jQ)$$

$$= 3(102 + j56.5)\ \text{KVA}$$

$$= (306 + j169.5)\ \text{KVA}$$

(b) Power factor =



$$\phi = 29^\circ$$

$$\therefore \cos\phi = 0.87 \cdot (\text{lag}).$$

(c) Can you calculate line current ?

⇓

Same as phase current in Wye-connected system.